Optical Properties of Snow Surfaces: Multiangular Photometric and Polarimetric Hyperspectral Measurements

Zhongqiu Sun, Di Wu, and Yunfeng Lv

***Abstract*— Understanding the optical properties of light reflected from snow surfaces is fundamental for remote sensing-based characterizations of snow properties and quantification of radiative transfer in the atmosphere–earth system. In theory, both intensity and polarization are required for describing the optical properties of light reflected from snow. However, thus far, a few studies have focused on the polarimetric properties of snow. In this study, we measured both multiangular photometric and polarimetric (based on the Stokes parameters) hyperspectral field results of snow with different properties (grain size, pollution levels, and microscopic surface roughness). By considering the absorption ratio of the polarizer, the intensity defined by photometric results [bidirectional reflectance factor (BRF) and hemispherical directional reflectance factor (HDRF)] was first confirmed to be similar to those derived from polarimetric measurements [*I* parameter reflectance factor (*I*pRF)] of snow surfaces. Then, the bidirectional polarized reflectance factor (BPRF) of snow, which is used to define the polarimetric properties of snow, was found to be useful for characterizing snow properties. Moreover, comparing ground measurements of snow BPRF with modeled results, we found that some existing BPRF models did not match well with the BPRF measured in the 2π space. These results suggest that more attention should be paid to the combination of *I*pRF (BRF or HDRF) and BPRF because polarimetric measurements can be used both for describing the optical properties of light reflected from snow and for quantifying snow properties.**

***Index Terms*— Bidirectional polarized reflectance factor (BPRF), multiangle, optical properties, photometric measurement, polarimetric measurement, snow.**

I. Introduction

QUANTIFYING the variation of solar radiation absorbed and reflected by the Earth's surface is crucial for understanding climate change [1], [2]. The amount of shortwave radiation absorbed depends on both the incident radiation and the surface reflection properties. The high albedo of snow is a key factor in the radiation balance of the Earth-atmosphere system for global climate, environmental change, and hydrological studies [3]–[6]. The spectral variation of snow is a function of several factors, such as the spectral and angular characteristics of solar incident radiation, the viewing-incident geometries, and the physical and chemical properties of snow itself [7]–[11]. Understanding the optical properties of light reflected from snow with different properties is important for characterizing its surface energy budget and estimating snowpack properties through remote sensing at various scales [12]–[16].

Photometric measurements, which are defined in terms of hyperspectral and angular reflectance factors [17]–[20], have been used to describe the optical properties of light reflected from snow. This is because the spectral reflectance factor depends on the snow properties, and snow is an anisotropic scatterer with a strong forward scattering behavior [7], [21]–[24]. Based on an in-depth understanding of the spectral and angular photometric properties of snow [25]–[27], ground, airborne, and spaceborne remote sensing techniques can be used to monitor the temporal and spatial variation of different types of snow at various scales [23], [28]–[36]. Photometric applications not only improve the representation of the radiative transfer between atmosphere and snow surfaces [22], [23], [37] but also demonstrate that optical remote sensing is useful for parameterizing snow properties [12], [16], [38], [39]. Theoretically, both photometry and polarimetry can be used to describe the optical properties of light [40]; yet, remote sensing studies have thus far focused mainly on the photometric properties and paid little attention to the polarimetric properties of targets [41].

Polarimetric measurements have the potential to characterize the land surfaces because polarization varies with the surface structural properties and can provide additional information for land cover characterization [40], [42]–[45]. An increasing body of evidence also confirmed that the polarimetric properties of snow at different viewing zenith angles and wavelengths are necessary for characterizing the optical properties of snow [46]–[49]. These investigations extended from ground and airborne to spaceborne measurements, deepening the understanding of snow's polarimetric properties at different scales [50]–[55]. Based on the measured and modeled polarimetric properties of snow, the main conclusions of these studies were that the degree of linear polarization (Dolp) of snow has the potential to characterize snow grain size and

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that the bidirectional polarized reflectance factor (BPRF) of snow is spectrally neutral [50], [51], [52], [54], [56]. Thus, consider that the polarimetric properties of snow improve the accuracy of snow grain size retrievals that are based solely on intensity measurements [50]–[52]. Moreover, it allows the use of the established methods for polarimetric retrievals of aerosol parameters in snow-covered regions [53], [56]. These results further indicate that information about snow surface polarimetric characteristics is needed to study the land–atmosphere system.

However, photometric and polarimetric measurements of snow have usually been performed with two independent methods: the former is directly using a photometer [7] and the latter is using a photometer with a polarizer [49], [52]. The different measurement methods complicate studies that require both photometric and polarimetric information to characterize the optical properties of snow. Moreover, the relationships between BPRF and snow properties (grain size, pollution content, and microscopic surface roughness) have not been investigated, limiting the extent to which polarimetric measurements can be used for the characterization of snow properties.

In theory, the Stokes parameters (I, Q, U, and V), which have been used to describe the polarimetric properties of natural surfaces [41], can help derive both photometric and polarimetric measurements of snow. The I parameter represents the intensity of snow, while Q and U characterize the polarization of snow. However, the I parameter and the polarized signals derived from the Stokes parameters have not yet been used to characterize snow properties. In this study, we compared photometric measurements [defined as the bidirectional reflectance factor (BRF) and hemispherical directional reflectance factor (HDRF)] and polarimetric measurements [defined as the I parameter reflectance factor (IpRF)] of snow in the field to confirm that the two methods derive similar estimates of the intensity of snow. Then, we analyzed the relationships between BPRF and snow properties (grain size, pollution, and surface microscopic roughness). Furthermore, we also used the BPRF of snow in the 2π space to evaluate the polarimetric model used in airborne and spaceborne polarization measurements.

II. Theory and Methods

A. Definition of the Photometric Properties of Snow

BRF and HDRF are usually used to characterize the photometric properties of snow under artificial light and sunlight conditions [10], [33], respectively. BRF is the ratio of the radiance reflected from the snow surface (dL_{snow}) to the radiance reflected from an ideal and diffuse reference surface ($dL_{\text{reference}}$) under the same incident-viewing geometry and measurement conditions [57]

$$\text{BRF} = \frac{dL_{\text{snow}}(\lambda, \theta_s, \varphi_s; \theta_v, \varphi_v)}{dL_{\text{reference}}(\lambda, \theta_s, \varphi_s; \theta_v, \varphi_v)}\rho_\lambda \tag{1}$$

where λ is the wavelength, θ_s is the source incident zenith angle, θ_v is the viewing zenith angle, φ_s is the source azimuth angle, and φ_v is the viewing azimuth angle. A Spectralon panel was used as the reference surface in this study, its reflectance, ρ_λ, provided by the manufacturer ranged from 0.9284 to 0.9954. The BRF of snow was measured using an artificial lamp as incident source.

The HDRF of snow is similar to the BRF and includes irradiance from the 2π space [57]

$$\text{HDRF} = \frac{dL_{\text{snow}}(\lambda, \theta_s, \varphi_s, 2\pi; \theta_v, \varphi_v)}{dL_{\text{reference}}(\lambda, \theta_s, \varphi_s, 2\pi; \theta_v, \varphi_v)}\rho_\lambda. \tag{2}$$

B. Definition of the Polarimetric Properties of Snow

The Stokes parameters (I, Q, U, and V) have been used to describe the polarimetric properties of snow: I describes the intensity of light reflected from snow, Q and U represent the magnitude and orientation of the semimajor axis of the polarization ellipse, respectively, and V is negligible after light has been reflected from the land surface [41]

$$I = (L_{0^\circ} + L_{45^\circ} + L_{90^\circ} + L_{135^\circ})/2 \tag{3}$$

$$Q = L_{0^\circ} - L_{90^\circ} \tag{4}$$

$$U = L_{45^\circ} - L_{135^\circ} \tag{5}$$

$$R_p = \sqrt{Q^2 + U^2} \tag{6}$$

where L_{0°, L_{45°, L_{90°, and L_{135° are the polarized reflected radiances from the snow surface in four different polarizer directions (0°, 45°, 90°, and 135°); the reference plane for the four polarizer directions is the meridional plane of the instrument. R_p is the polarized radiance reflected from snow. BPRF is used to describe the polarization of snow. BPRF is the ratio of the polarized radiance reflected ($dR_{p,\text{snow}}$) from snow to the radiance reflected ($dL_{\text{reference}}$) from the reference surface (the Spectralon panel) under the same incident-viewing geometry and measurement conditions [45], [58]

$$\text{BPRF} = \frac{dR_{p,\text{snow}}(\lambda, \theta_s, \varphi_s; \theta_v, \varphi_v)}{dL_{\text{reference}}(\lambda, \theta_s, \varphi_s; \theta_v, \varphi_v)}\rho_\lambda. \tag{7}$$

When I parameter is used to describe the photometric properties of light reflected from snow under lamp and sunlight conditions, $I\text{pRF}_{\text{lamp}}$ (or $I\text{pRF}_{\text{sun}}$), which are the ratio of half of the sum of polarized radiance ($dI_{\text{snow,lamp}}$ or $dI_{\text{snow,sun}}$) in four polarizer directions [(3)] to the radiance reflected ($dL_{\text{reference,lamp}}$ or $dL_{\text{reference,sun}}$) from the reference surface (the Spectralon panel) under the same incident-viewing geometry and measurement conditions [58], is calculated as

$$I\text{pRF}_{\text{lamp}} = \frac{dI_{\text{snow,lamp}}(\lambda, \theta_s, \varphi_s; \theta_v, \varphi_v)}{dL_{\text{reference,lamp}}(\lambda, \theta_s, \varphi_s; \theta_v, \varphi_v)}\rho_\lambda \tag{8-1}$$

$$I\text{pRF}_{\text{sun}} = \frac{dI_{\text{snow,sun}}(\lambda, \theta_s, \varphi_s, 2\pi; \theta_v, \varphi_v)}{dL_{\text{reference,sun}}(\lambda, \theta_s, \varphi_s, 2\pi; \theta_v, \varphi_v)}\rho_\lambda. \tag{8-2}$$

$I\text{pRF}_{\text{lamp}}$ represents the results measured using the lamp and $I\text{pRF}_{\text{sun}}$ represents the results measured under the sunlight. Theoretically, $I\text{pRF}_{\text{lamp}}$ or $I\text{pRF}_{\text{sun}}$ is the same as the BRF or HDRF of snow. However, the absorption ratio of the polarizer reduces the polarized radiance at different polarizer directions (0°, 45°, 90°, and 135°) measured by the spectrometer, causing the BRF or HDRF to be larger than $I\text{pRF}_{\text{lamp}}$ or $I\text{pRF}_{\text{sun}}$ at each wavelength. Under the consideration of the absorption

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ratio of the polarizer, the I parameter in (3) can be redefined as follows:

$$I = \left(\frac{L_{0^\circ}}{1 - abs_p} + \frac{L_{45^\circ}}{1 - abs_p} + \frac{L_{90^\circ}}{1 - abs_p} + \frac{L_{135^\circ}}{1 - abs_p} \right) / 2 \tag{9}$$

where abs_p is the absorption ratio of the polarizer and can be calculated using the difference between the radiance of the Spectralon panel measured with a polarizer and without a polarizer. A detailed description can be found in a previous study [58]. A calcite Glan–Thompson prism with an effectivity from 350 to 2300 nm was placed in front of the optic fiber of the ASD spectrometer, and we measured the polarized radiance reflected from snow surfaces at different polarizer directions from 0° to 360° in 1° increments.

Hereafter, the IpRF of snow represents the results with consideration of abs_p. In this study, the absolute value of the relative difference (aRD) was used to assess the agreement between $I\text{pRF}_{\text{lamp}}$ (or $I\text{pRF}_{\text{sun}}$) and BRF (or HDRF)

$$\text{aRD} = \left| \frac{I\text{pRF}_{\text{lamp}} - \text{BRF}}{\text{BRF}} \right| \text{or} \left| \frac{I\text{pRF}_{\text{sun}} - \text{HDRF}}{\text{HDRF}} \right|. \tag{10}$$

The small aRD values correspond to the good agreement between $I\text{pRF}_{\text{lamp}}$ (or $I\text{pRF}_{\text{sun}}$) and BRF (or HDRF), which indicates that the BRF (or HDRF) can be replaced by $I\text{pRF}_{\text{lamp}}$ (or $I\text{pRF}_{\text{sun}}$) with different properties.

C. Photometric and Polarimetric Models

In this study, the photometric properties of snow were modeled using the asymptotic radiative transfer (ART) model [27], [28], which was used to simulate the spectral and angular reflection of snow in ground and spaceborne-based studies [27], [32]. The description of the ART model is as follows:

$$\begin{aligned} &R(\theta_s, \varphi_s; \theta_v, \varphi_v) \\ &\quad = R_0(\theta_s, \varphi_s; \theta_v, \varphi_v) \\ &\qquad \cdot \exp[-\alpha K_0(\cos\theta_s) K_0(\cos\theta_v) / R_0(\theta_s, \varphi_s; \theta_v, \varphi_v)] \end{aligned} \tag{11}$$

$$\begin{aligned} &R_0(\theta_s, \varphi_s; \theta_v, \varphi_v) \\ &\quad = \frac{a + b(\cos\theta_s + \cos\theta_v) + c\cos\theta_s\cos\theta_v + p(\gamma)}{4(\cos\theta_s + \cos\theta_v)} \end{aligned} \tag{12}$$

$$K_0(x) = \frac{3}{7}(1 + 2x) \tag{13}$$

$$\cos\gamma = -\cos\theta_s\cos\theta_v - \sin\theta_s\sin\theta_v\cos(\varphi_s - \varphi_v) \tag{14}$$

where γ is the scattering angle (in degrees), the azimuthal angle of 180° represents the forward scattering direction, $R(\theta_s, \varphi_s; \theta_v, \varphi_v)$ corresponds to the BRF or HDRF of snow, $a = 1.247$, $b = 1.186$, and $c = 5.157$ [27]

$$p(\gamma) = 11.1\exp(-0.087\gamma) + 1.1\exp(-0.014\gamma) \tag{15}$$

$$\alpha = \sqrt{\eta L} \text{ and } \eta = 4\pi(\chi + M)/\lambda \tag{16}$$

where χ is the imaginary part of ice refractive index (the values from 400 to 2500 nm are referred in [59]), L is used to parameterize snow grain size, M is proportional to the concentration of impurities in the snow, and λ is the wavelength.

Three polarimetric models were used to calculate the BPRF of snow in this study. The first polarimetric model is a linear one-parameter polarized reflectance model (defined as M-model in this study) that has been applied to model the polarized reflectance factors of land surfaces [54]

$$\text{BPRF}(\theta_s, \varphi_s; \theta_v, \varphi_v) = \frac{C\exp(-\tan\alpha)\exp(-\upsilon)F(\alpha, n)}{4(\cos\theta_s + \cos\theta_v)} \tag{17}$$

where C is the free linear parameter of this model, υ is related to the normalized difference vegetation index (NDVI) [54], and $F_p(\alpha, n)$ is the Fresnel equation, which is a function of the refractive index n and the incidence angle α. The incidence angle (α) is half the phase angle [which is supplementary to the scattering angle (γ) in (14)].

The second polarimetric model is the Nadal–Breon polarized reflectance model (NB-model) [60], which has been used to model the polarized reflectance factor of land surface and can be written as follows:

$$\text{BPRF}(\theta_s, \varphi_s; \theta_v, \varphi_v) = D\left[1 - \exp\left(-B\frac{F_p(\alpha, n)}{\cos\theta_s + \cos\theta_v}\right)\right] \tag{18}$$

where D and B are the parameters of the NB model.

The third polarimetric model is a modified Fresnel model for land surface polarized reflectance factors (L-model) [61], which has been used to model the polarization of the earth surface from airborne measurements

$$\text{BPRF}(\theta_s, \varphi_s; \theta_v, \varphi_v) = \frac{H\pi F(\alpha, n)}{4\cos\beta(\cos\theta_s + \cos\theta_v)} f(\sigma, \beta) f_{sh}(\gamma) \tag{19}$$

$$f(\sigma, \beta) = \frac{1}{\pi\cos^3\beta 2\sigma^2}\exp\left(-\frac{\tan^2\beta}{2\sigma^2}\right) \tag{20}$$

$$f_{sh}(\gamma) = \left(\frac{1 + \cos k_s(\pi - \gamma)}{2}\right)^3 \tag{21}$$

$$\cos\beta = \frac{\cos\theta_v + \cos\theta_s}{2\cos\alpha}. \tag{22}$$

There are three parameters in this model: H, σ, and k_s. σ^2 represents the mean square facets slope, $f(\sigma, \beta)$ describes the distribution of the facets (assumed to follow as a Gaussian distribution), and $f_{sh}(\gamma)$ is a shadowing function [61].

The three polarimetric models have been used to calculate the polarization of snow from spaceborne measurements [55]. However, they have not been evaluated by the BPRF of snow from ground measurements, which represents the physical polarimetric properties of snow. In this study, the measured photometric and polarimetric results were used to invert the model parameters based on the "Isqnonlin" function in MATLAB. The square root mean square errors was used in the optimization performance

$$\text{RMSE} = \sqrt{\frac{\sum_{k=1}^{n}(R_m - R)^2}{N_f}} \tag{23}$$

where R_m and R are the measured and modeled results in the same viewing-incident geometry, and N_f is the number

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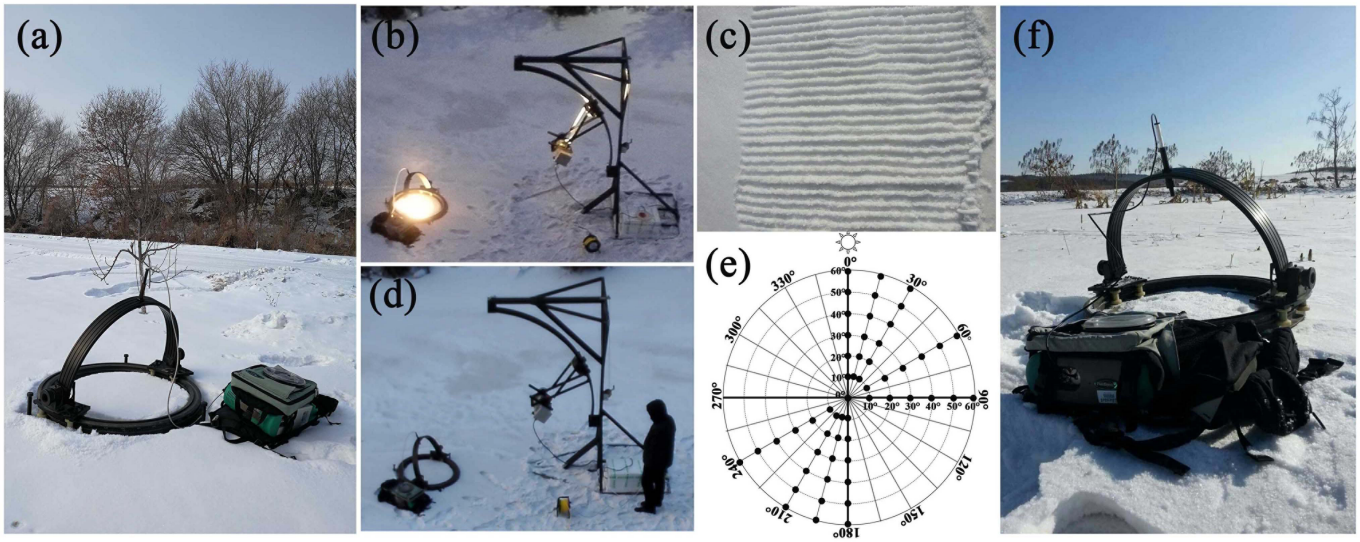


Fig. 1. Measurements at Site 1 (a) and (f) under the sunlight conditions. Measurements at Site 2 (b) and (d) under artificial illumination. (c) Artificial rough snow surface. (e) Schematic measurements under the artificial illumination; the soil lines with circular soil points represent the viewing directions, 0° refers to the backward scattering direction in the principal plane. In the field (a) and (f), the measurements were only performed in the principal plane.

TABLE I

SITE, INCIDENT SOURCE, SNOW PROPERTIES, AND AIR TEMPERATURE. DEPOSITED-BC REPRESENTS THE BC DEPOSITED ON THE SNOW

Location	Date	Incident source	Snow type	Incident angle	Temperature	Depth
Site 1	2015-02-07	Sun	Old snow (5 day)	60°	−17.5 °C	0.35 m
Site 2	2017-02-06	Lamp	Old snow (3-4 day)	65°	-17.5 °C	0.45 m
Site 2	2017-02-08	Lamp	Old snow (5-6 day)	65°	−17 °C	0.45 m
			Rough snow surface		−17.5 °C	0.45 m
Site 2	2017-02-10	Lamp	New snow (1 day)	65°	−18 °C	0.3 m
	2017-02-12	Lamp	> 0.9 mm old snow (3 day)		−18 °C	0.3 m
			0.45-0.9 mm old snow (3 day)		−18 °C	0.3 m
	2018-02-04	Lamp	Old snow (5-6 day) (559 ng/g)	65°	−17.5	0.35 m
			Deposited-BC (925 ng/g)		−18 °C	0.35 m
			Deposited-BC (1356 ng/g)		−17 °C	0.35 m
Site 1	2018-02-06	Sun	Old snow (7-8 day) (472 ng/g)	63°-65°	−10.5 °C	0.4 m

of degrees of freedom, which is the difference between the number of measurements and the number of model parameters.

D. Site and Sample Descriptions

The photometric and polarimetric measurements of snow were performed at two sites in northern China. Site 1 is located in a cultivated field near Changchun City [43.7315 °N and 125.4621 °E, Fig. 1(a) and (f)], Site 2 is located in a grass playground at Changchun Normal University [43.9155 °N and 125.3949 °E, Fig. 1(b) and (d)]. Snow with the different properties (grain size, pollution level, and macroscopic surface roughness, as shown in Table I) was not only used to illustrate the similarity between $I\text{pRF}_{\text{lamp}}$ (or $I\text{pRF}_{\text{sun}}$) and BRF (or HDRF) under different measurement conditions but was also used to relate to the BPRF of snow.

Snow samples with different grain sizes were prepared using two sieves with 0.9 and 0.45 mm, producing three different grain size distributions: >0.9 mm, 0.45–0.9 mm, and <0.45 mm. To avoid the influence of pollution in snow with small grains (<0.45 mm) on the spectral reflectance factor, only snow with grain size >0.9 and 0.45–0.9 mm on February 12, 2017 and new snow on February 10, 2017 at Site 2 (Table I) were selected to investigate the effect of grain size on the photometric and polarimetric properties of snow.

The polluted snow samples had different black carbon (BC) concentrations. BC was collected from snow samples near the two sites at depths less than 5 cm. After melting the snow samples, the impurities were separated with a 0.4-μm nuclepore filter. We assumed that the impurities were BC based on previous snow studies done near our sampling sites [62]–[64]. The collected BC then was deposited on the snow surfaces to yield snow with different BC contents at Site 2 on 4 February 2018 (see Table I). Because the measurements were taken under stable and cold conditions (from −18 ° to −17 ° in Table I), we assumed that the BC content was the only factor influencing the optical properties of light reflected from these snow surfaces.

An artificial rough snow surface was created on a square of 0.5 m × 0.5 m at Site 2 on February 8, 2017 by creating a series of 1.5 cm deep and 1-cm-wide parallel ridges [Fig. 1(c)], spaced about 1 cm apart. Researchers have used similar artificial rough shapes to investigate the effect of microscopic surface roughness on snow directional reflectance and

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albedo [65]–[67]. The photometric and polarimetric measurements from the rough snow surface were compared with those from the flat snow surface at Site 2 on February 8, 2017 in Table I. The roughness axis (defined as the direction along the ridges of roughness) was perpendicular to the incident zenith direction in this study.

E. Photometric and Polarimetric Measurements

The artificial illumination system adopted in this study (Lamp in Table I) consisted of a 1000-W halogen lamp (Oriel, Newport Corporation, Irvine, CA, USA), a 90° arc with 1.5 m radius, and an OFF-axis parabolic mirror [see Fig. 1(b) and (d)]. An OFF-axis parabolic mirror was attached to the arc to imitate sunlight at different incident zenith angles with an accuracy of 0.5° and was used to collimate the light from the halogen lamp [32]. Our artificial illumination system features are similar to those of the Finnish Geodetic Institute Field Goniospectrometer, which allows researchers to measure both photometric and polarimetric properties of snow surfaces [18], [33], [49]. The angular photopolarimetric properties of snow were measured using a goniometer composed of a rotating arm and a circular base with a 0.4 m radius [see Fig. 1(a) and (f)]

Theoretically, any point on the hemisphere of the snow samples can be measured using this goniometer. Due to the goniometer's structural limitation, 10° is the smallest phase angle that can be obtained during the variation of the viewing direction. Under artificial illumination, both photometric and polarimetric properties of snow were measured in 53 different directions, as shown in Fig. 1(e). Under the solar incident light, we measured the photometric and polarimetric properties of snow in ten directions in the principal plane [corresponding to the 0°–180° line in Fig. 1(e)]. The field of view of the sensor was 6° for both photometric and polarimetric measurements, and the measured footprint was from 4.2 cm (a circular area) to 8.4 cm (an elliptical area) when the viewing zenith angle changed from 0° to 60°. We assumed that the snow grain size did not influence the optical properties of snow with different BC contents and snow with roughness before February 4, 2018 because the crystal growth rate was very slow at temperatures lower than −16 ° [68] (see Table I) and the BC was deposited on the same snow surface. We also assumed that the snow samples' properties did not change during the course of the photometric and polarimetric measurements on February 6, 2018, when the temperature changed from −10.5 ° to −11 °, as each measurement in the principal plane took about 10 min.

III. RESULTS

A. Comparison Between IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) of Snow With Different Properties

The spectral and angular reflection of snow (BRF or HDRF) is directly influenced by the change of snow properties, such as grain size, pollution level, and microscopic surface roughness [2], [4], [9], [10]. If the IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) is similar to the BRF (or HDRF) of snow, the polarimetric measurements based on the Stokes parameters can characterize both the intensity and polarization of snow with different properties. Here, the comparison between IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) of snow with different properties was used to indicate that the reflectance factor derived from the polarimetric measurements was similar to that derived from photometric measurements.

The spectral BRF (or HDRF) and IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) of snow with different properties and the absolute value of the relative difference [aRD in (10)] between them are shown in Fig. 2. The IpRF$_{\mathrm{lamp}}$ and IpRF$_{\mathrm{sun}}$ have a spectral variation similar to that of the BRF and HDRF: the reflectance factor of a flat [Fig. 1(a) and (b); (m) and (n)] and a rough snow surface [Fig. 1(j) and (k)] increased in the forward scattering direction, the reflectance factor decreased with the increase of BC [Fig. 1(d) and (e)], and the reflectance factor increased with the decrease of grain size [Fig. 1(g) and (h)]. Moreover, the maximum aRD of snow with different parameters was not larger than 0.06 in the wavelength range from 400 to 1500 nm [Fig. 1(c), (f), (i), (l), and (o)], and the average aRD across this wavelength range was smaller than 0.035 for snow with different parameters, indicating that the average difference between IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) over the range from 400 to 1500 nm was smaller than 3.5%.

In the wavelength range from 1500 to 2500 nm with low reflectance factors, the aRD was relatively large at each wavelength; the maximum aRD was larger than 0.5 at wavelengths greater than 2000 nm. In fact, the difference between IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) over all wavelengths from 400 to 2500 nm was small. For example, snow with BC in the concentration of 1356 ng/g had IpRF$_{\mathrm{lamp}}$ and BRF of 0.503 and 0.518 at 900 nm; the difference between IpRF$_{\mathrm{lamp}}$ and BRF was 0.015, which led to a value of aRD was 0.029, but at 2170 nm, the IpRF$_{\mathrm{lamp}}$ and BRF were 0.0305 and 0.0381, respectively, and the difference between IpRF and BRF was 0.0076, which led to the aRD value of 0.20. This means that low reflectance factors corresponded to a relatively large aRD. These results in Fig. 2 confirmed the spectral consistency between IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) at wavelengths ranging from 400 to 2500 nm.

There was good agreement between spectral IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) in Fig. 2. To further confirm that IpRF$_{\mathrm{lamp}}$ (or IpRF$_{\mathrm{sun}}$) and BRF (or HDRF) are similar at different viewing zenith angles, we derived the distribution of aRD of snow with varying properties at 445, 670, and 1030 nm [Figs. 3–5]. The wavelengths of 445, 670, and 1030 nm were selected to represent the angular properties of light reflected from snow surfaces because airborne and spaceborne sensors use these to measure the multiangular photometric and polarimetric properties of snow [35], [69], [70]; in addition, the reflectance factors at the three wavelengths are also sensitive to snow properties, such as changes in BC content and grain size [16], [71].

In Fig. 3, it is found that the angular distribution of both BRF and IpRF$_{\mathrm{lamp}}$ of snow with a flat surface (on February 8, 2017) at different wavelengths was anisotropic; the maximum values were located in the forward scattering directions, and a relatively flat distribution pattern was found near the backward scattering directions. The angular distribu-

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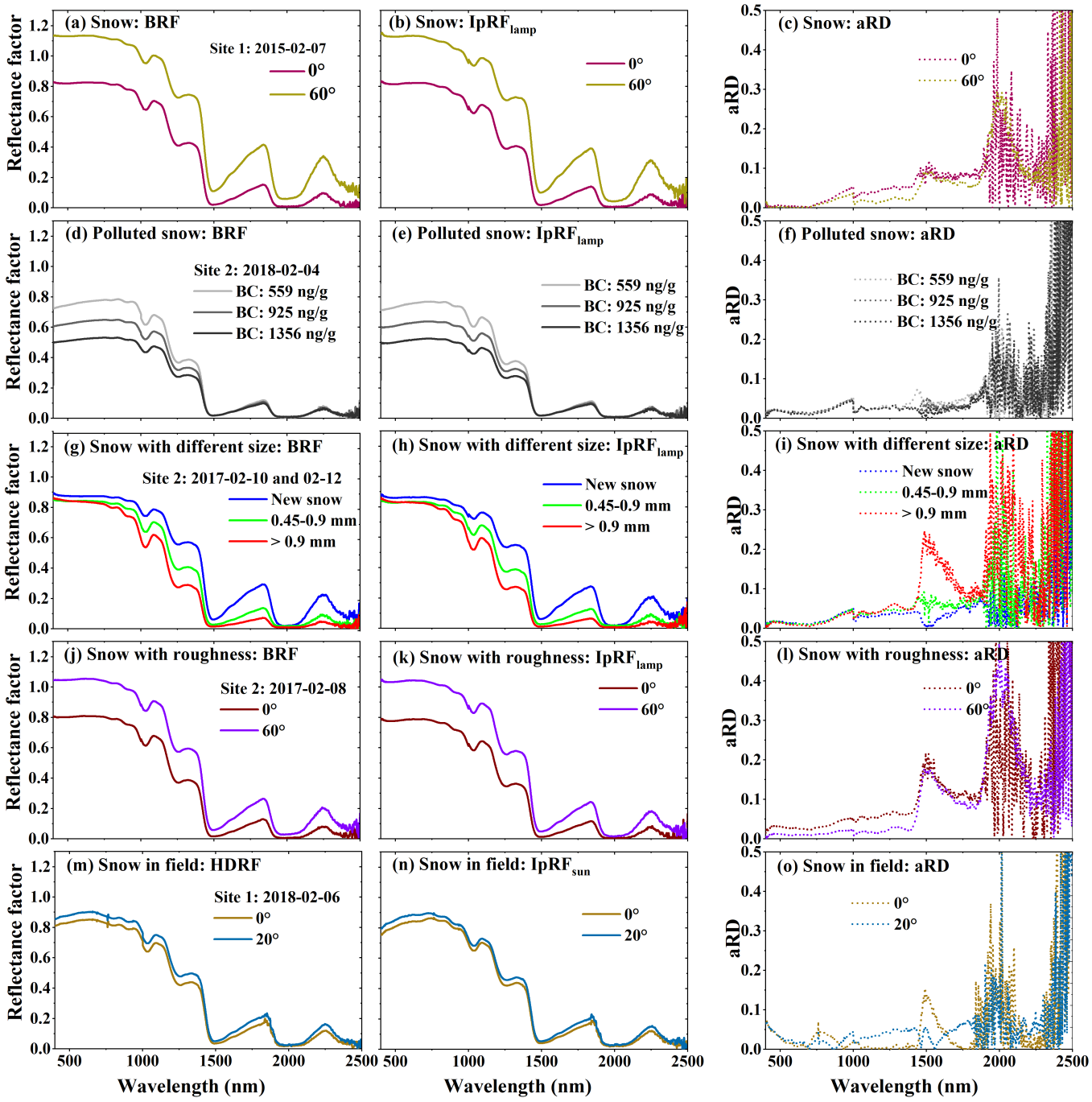


Fig. 2. BRF and HDRF of snow [left column: (a), (d), (g), (j), and (m)], the IpRF$_{lamp}$ and IpRF$_{sun}$ of snow [central column: (b), (e), (h), (k), and (n)] with different parameters, and the aRD [right column: (c), (f), (I), (l), and (o)]. From top to bottom, the reflectance factors and aRD correspond to old snow with a flat surface at 0° (nadir) and 60° (forward scattering direction), snow with different BC content at nadir, snow with different grain size at nadir, snow with roughness at 0° (nadir) and 60° (forward scattering direction), and old snow with flat surface at 0° (nadir) and 20° (forward scattering direction).

tion of snow was due to the forward scattering properties of snow, which has been observed and discussed in the previous studies [10], [20], [36], [72]. The aRD at 445, 670, and 1030 nm in Fig. 3(c), (f), and (i) was less than 0.06 in any viewing directions, and the average aRD at each selected wavelength over all viewing direction was less than 0.028. These results indicate that the IpRF$_{lamp}$ was similar to the BRF of a flat snow surface when the measurements were performed at different viewing zenith angles.

The effect of microscopic roughness on the distribution of BRF and IpRF$_{lamp}$ of snow (on February 8, 2017) is shown in the left and central columns in Fig. 4. It is found that the roughness alters the reflection distribution of snow. As the incident azimuth was perpendicular to the roughness axis, the BRF and IpRF$_{lamp}$ of a rough snow surface were dramatically enhanced in the backward scattering directions and strongly reduced in the forward scattering directions at different wavelengths when compared with the distribution

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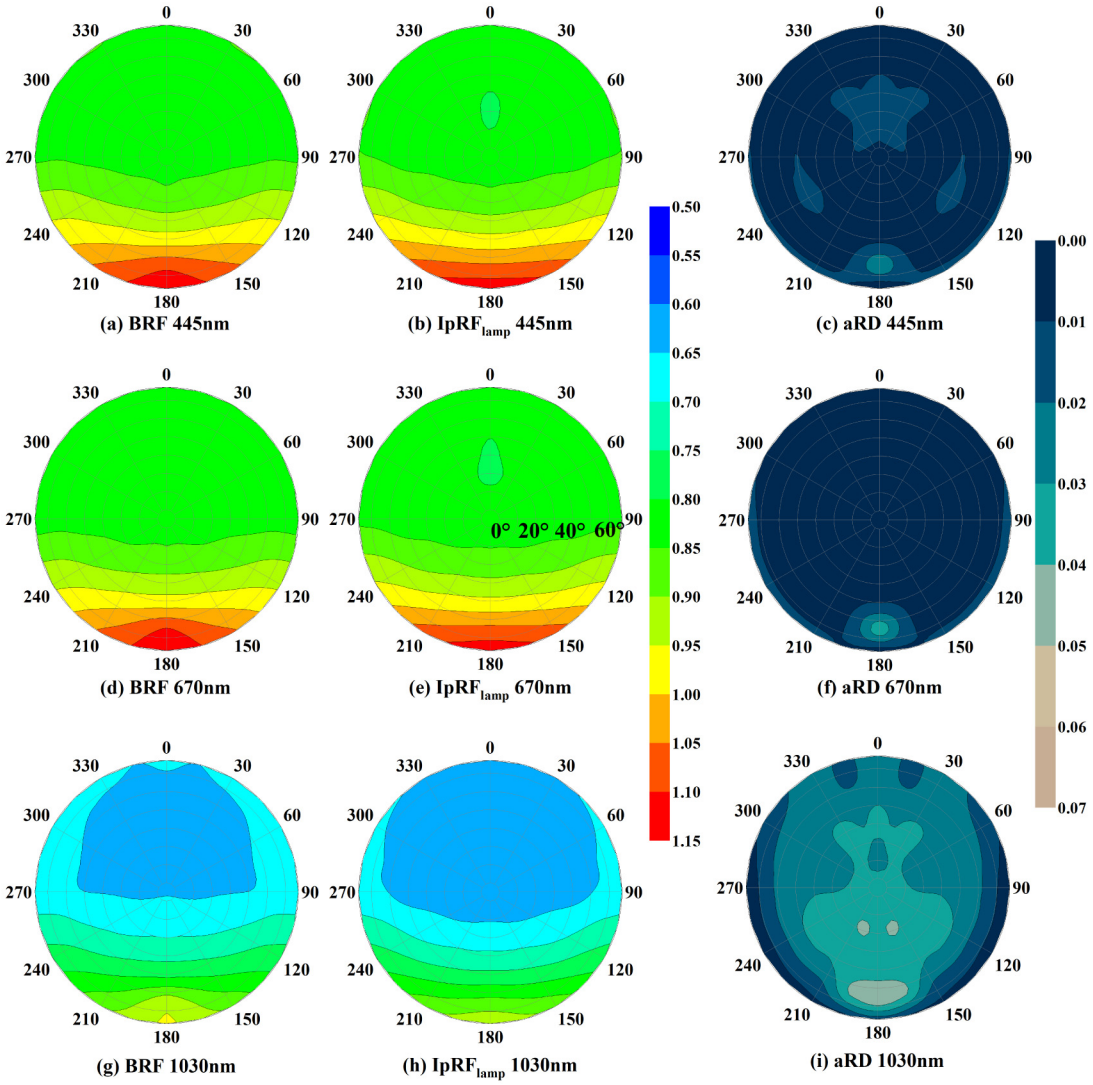


Fig. 3. Distribution of BRF (left column), IpRF$_{lamp}$ (central column), and aRD (right column) at 445 nm [first line: (a)–(c)], 670 nm [second line: (d)–(f)], and 1030 nm [third line: (g)–(i)] of a flat surface of old snow measured on February 8, 2017 at Site 2.

of BRF and IpRF$_{lamp}$ of snow with a flat surface in Fig. 3. The reduction in reflectance factors in the forward scattering directions was due to the shadows caused by the roughness, and the increment of reflectance factors in the backward scattering directions was caused by changing the effective incident zenith angle and enhancing the multiple scattering between adjacent roughness [65], [66], [73], [74]. Although the magnitude of the decrease and increase of reflectance factors depends on the incident zenith angle, viewing zenith angle, and the size and density of roughness [66], [75], we will not discuss the effects of these conditions on the distribution of the reflection of a rough snow surface. We focus on the investigation of the agreement between BRF and IpRF$_{lamp}$ of rough snow. In the right column of Fig. 4, it is found that the aRD is not larger than 0.06 in the three selected wavelengths in any viewing zenith angle, and the average aRD of each selected wavelength over all the viewing directions is less than 0.033. The good agreement between BRF and IpRF$_{lamp}$ of rough snow further indicates that BRF can be replaced by the IpRF$_{lamp}$ derived from polarimetric measurements.

In Fig. 5, the HDRF and IpRF$_{sun}$ of snow with a flat surface measured at different viewing zenith angles in the principal plane were used to further confirm the small difference between them. The aRD was less than 0.06 at each viewing zenith angle and the average aRD of each wavelength over all the viewing directions was less than 0.045 [see Fig. 5(c)]. In summary, the average aRD between spectral IpRF$_{lamp}$ (or IpRF$_{sun}$) and BRF (or HDRF) in the wavelength range from 400 to 1500 nm was not larger than 0.035 for all snow samples, and the average aRD between angular IpRF$_{lamp}$ (or IpRF$_{sun}$) and BRF (or HDRF) in each selected wavelength (445, 670, and 1030 nm) was not larger than 0.045. Considering that the average uncertainty of the photometric measurements was about 1% and the average uncertainty of the polarimetric measurements was about 2% using the measurement system in this study [32], the IpRF$_{lamp}$ (or IpRF$_{sun}$)

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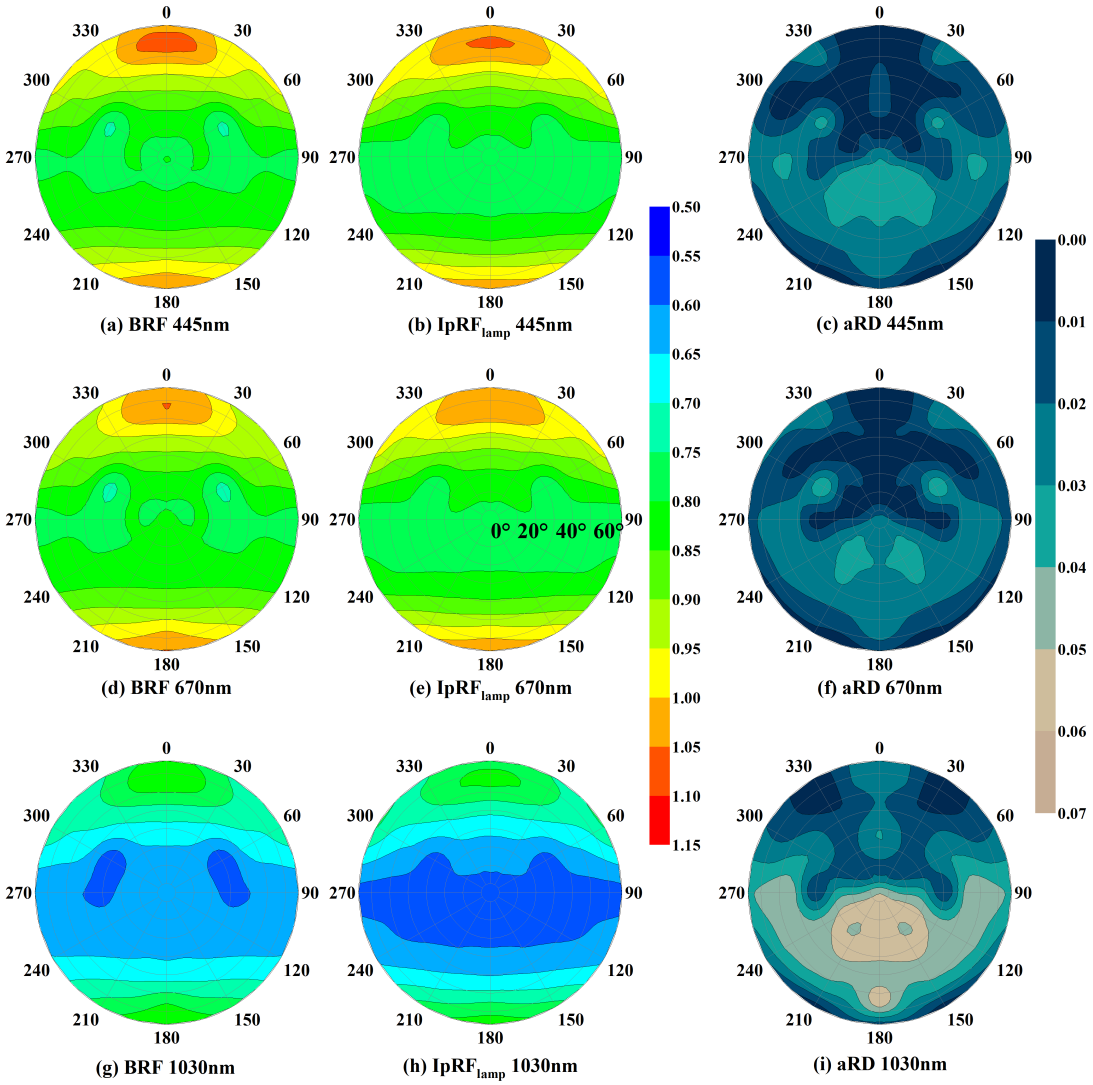


Fig. 4. Distribution of BRF (left column), IpRF$_{lamp}$ (central column), and aRD (right column) at 445 nm [first line: (a)–(c)], 670 nm [second line: (d)–(f)], and 1030 nm [third line: (g)–(i)] of snow with microscopic roughness measured on February 8, 2017 at Site 2. The roughness axis was perpendicular to the incident azimuth; 180° corresponds to the forward scattering direction.

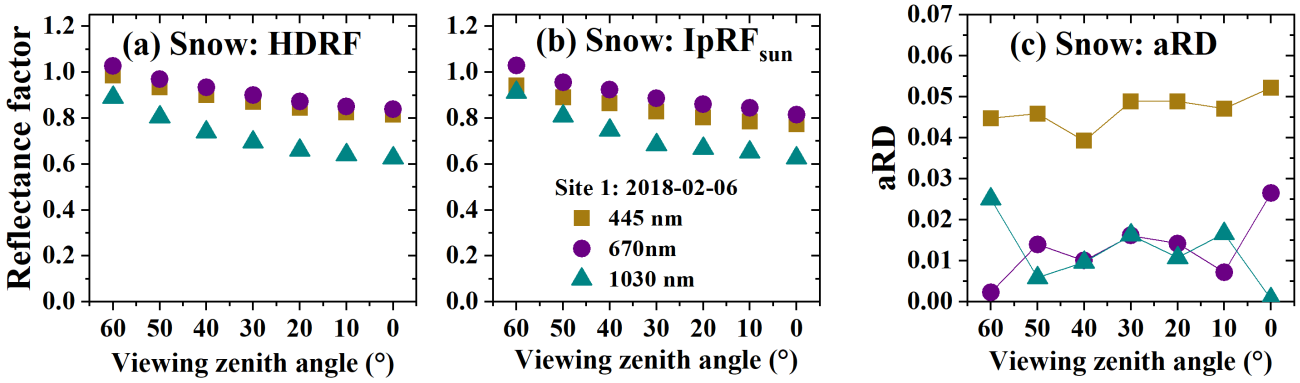


Fig. 5. Angular distribution of (a) HDRF, (b) IpRF$_{sun}$, and (c) aRD of snow under sunlight in the forward scattering directions in the principal plane. The results were measured in February 6, 2018 at Site 1.

can be confirmed to be similar to the BRF (or HDRF) of snow with different properties.

In the right column in Figs. 3 and 4, the aRD at 1030 nm is larger than the aRD at 445 and 670 nm, and the maximum aRD is usually found in the forward scattering directions. This may be because we first performed the photometric measurements at different viewing directions, which took 40–50 min; then, we began the polarimetric measurements, which took 1.5 h.

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The snow surface temperature increased after 40–50 min of exposure to artificial light when we performed the polarimetric measurements. Under the "warmer" conditions, the snow grain size slightly increased [68]. Because the reflectance factor at 1030 nm is more sensitive to grain size than the reflectance factor at 445 and 670 nm, there was a relatively large aRD between BRF and $I\text{pRF}_{\text{lamp}}$ at 1030 nm. Under the "warmer" conditions caused by the long-time exposure to light, the snow grain shape also changed [68]. Because the distribution of the reflectance factor is very sensitive to snow grain shape, especially in the forward scattering directions [4], [72], the maximum aRD was usually found in the forward scattering directions.

For the aRD shown in Fig. 5(c), the aRD at 670 nm and 1030 were smaller than at 445 nm. During the field measurements, both the photometric and polarimetric measurements were performed in the principal plane under a clear sky; the measurements took 16 min, and we assumed that the snow properties did not change within this time under natural illumination conditions. However, the polarization of a clear sky may influence the polarimetric measurements of snow surfaces at short wavelengths in the visible bands [76], which may lead to a relatively large aRD at 445 nm. In order to shorten the measurement time in this study, we did not subtract the polarization of diffuse radiation from the snow surface as previous studies [50], [77]. In future studies, the consistency between the HDRF and $I\text{pRF}_{\text{sun}}$ of snow under different sky conditions should be further confirmed. Overall, the similar spectral and angular reflection characteristics of snow with different properties derived from the photometric and polarimetric measurements indicate that $I\text{pRF}_{\text{lamp}}$ (or $I\text{pRF}_{\text{sun}}$) can replace the BRF (or HDRF) under artificial and sunlight conditions, respectively.

B. Polarization of Snow With Different Properties

The spectral and angular BRF and HDRF have been used to represent the photometric properties of light reflected from snow surfaces with different properties. A complete description of snow's optical properties requires an understanding of the effect of snow properties on the spectral and angular characters of the BPRF [40]. Previous studies have found that the BPRF of land surfaces is spectrally neutral because the specular reflection or single scattering is the derivation of polarization [54], [55], [60]. Fig. 6 shows that the BPRF of flat snow surfaces in the range from 400 to 1000 nm is weakly wavelength-dependent under both artificial light and sunlight. Moreover, the spectral variation of BPRF [Fig. 6(b)] of snow is found around 670 nm under the sunlight condition.

In fact, the polarized reflectance factor of snow measured by the airborne polarimetric sensor (RSP: Research Scanning Polarimeter) had the similar spectral variation as shown in Fig. 6(b), and the relatively large BPRF was found at 670 nm when the viewing zenith angle increased [53], [56]. We cannot explain the spectral variation of BPRF in Fig. 6(b) in this study. We suspect that the variation in BPRF under the sunlight condition may be because we did not subtract the effect of sky diffuse radiation on the polarization of snow as

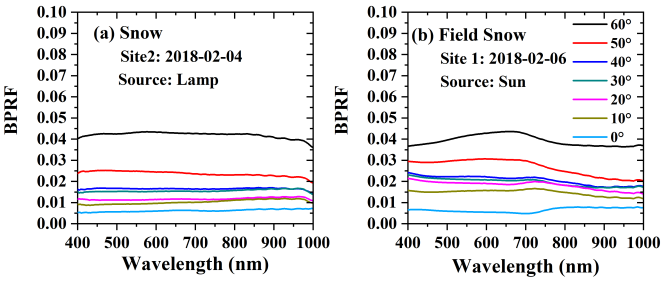


Fig. 6. Spectral BPRF of a flat snow surface under (a) artificial light and (b) sunlight. The measurements were performed at different viewing zenith angles (from 0° to 60°) in the forward scattering direction in the principal plane. Because of the polarization sensitivity of the spectrometer [49], we used the BPRF in the range of 400–1000 nm.

the previous studies [49], [50]. BPRF in the wavelength range from 400 to 1000 nm is selected to represent the polarimetric properties of snow because the ASD spectrometer is sensitive to the polarization in the wavelength range from 1000 to 2500 nm [49], [58], [77]. Some irregular noise leaps caused by the polarization sensitivity of the ASD spectrometer cannot be used to explain the polarimetric properties of snow with different properties [49], [58].

Although several studies investigated the spectral polarization properties of snow, which is usually defined by the degree of linear polarization (Dolp) [49], [50], [52], few focused on the spectral BPRF of snow with different properties. Moreover, the increase in BPRF from 0° to 60° in the forward scattering direction (Fig. 6) can be explained by the Fresnel equations. Highly polarized reflectance is found at large viewing zenith angles [60]. In this section, the focus is on the effects of grain size, pollution level, and microscopic roughness on the BPRF of snow.

1) Grain Size Effects: The spectral BPRF of snow with different grain sizes measured at different viewing zenith angles in the forward scattering directions in the principal plane is shown in Fig. 7(a)–(c): new snow, 0.45–0.9 mm, and >0.9 mm, respectively. There was a weak wavelength dependence of the BPRF of snow with different grain sizes at different viewing zenith angles [see Fig. 7(a)–(c)]. Notably, the BPRF increased with the decrease in snow grain size, such as the BPRF of new snow that had the highest BPRF at each viewing zenith angle, followed by the snow with 0.45–0.9 mm and >0.9 mm [see Fig. 7(a)–(c)]. Moreover, the distribution of BPRF of snow with different grain sizes at 670 nm [Fig. 7(d)–(f)] indicated that the decrease in grain size not only increased the BPRF of snow but also improved the anisotropy of BPRF of snow. Based on the geometrical optical method, researchers concluded that the particulate media with small particles was brighter than that with large particles [72], [78], and the amount of single scattering (or first order of scattering) of media with small particles was more than that of media with large particles [78], [79]. The BPRF is derived from the single scattering light reflected from snow surfaces [41]; thus, the smaller the grain size, the greater the single scattering, and the larger the BPRF (see Fig. 7). However, we cannot explain that the BPRF decreases with the increase in grain size in the backward scattering directions

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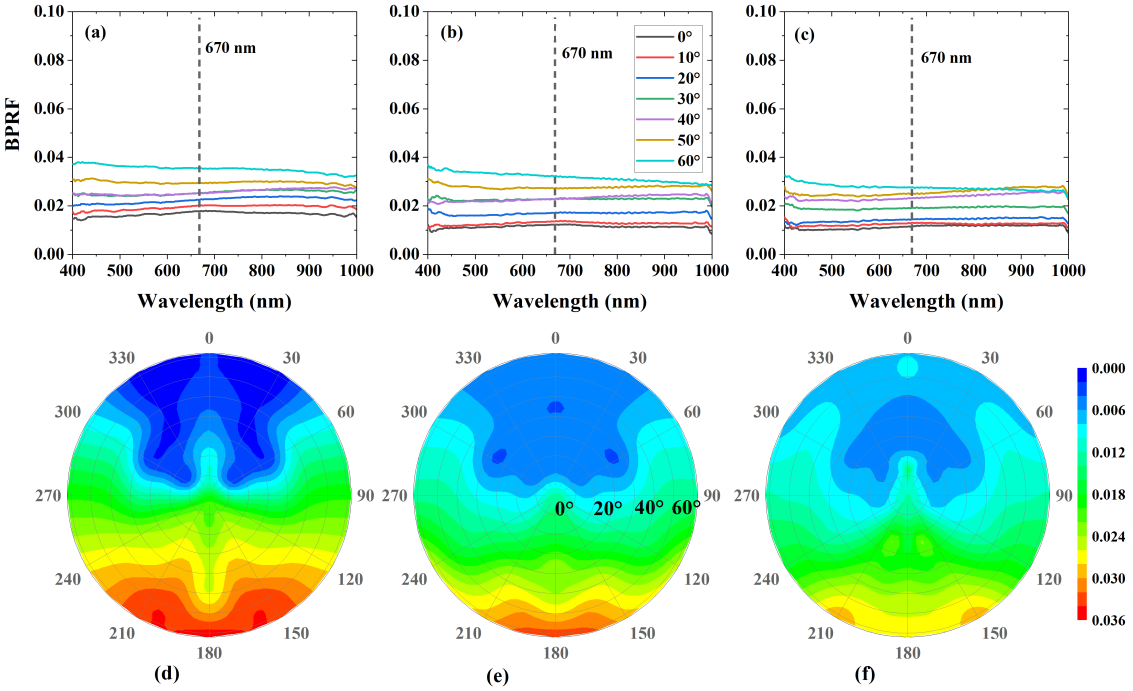


Fig. 7. Spectral BPRF of snow samples measured at Site 2 on February 10 and 12, 2017: (a) new snow, (b) 0.45–0.9 mm, and (c) >0.9 mm. New snow was used to represent snow with a small grain size. The distribution of BPRF at 670 nm for (d) new snow, (e) 0.45–0.9 mm, and (f) >0.9 mm.

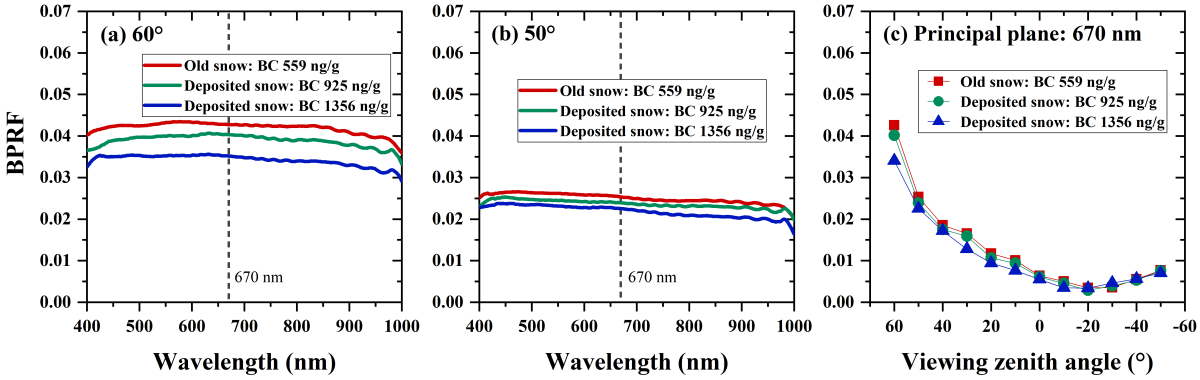


Fig. 8. Spectral BPRF of snow with different BC contents at (a) 60° and (b) 50° viewing zenith angle in the forward scattering direction. The distribution of BPRF at 670 nm (c) for snow with different BC contents from 0° to 60° in the principal plane, "−" represents the backward scattering direction. The snow samples were measured on February 04, 2018 at Site 2 under artificial illumination.

because several physical mechanisms control the polarization in these directions [50], [80]. More measurements should be performed to deepen our understanding of this variation in future studies.

2) Pollution Effects: The polarimetric properties of snow with different BC contents were measured at different viewing zenith angles in the principal plane. The spectral BPRF of snow with different BC contents in the viewing zenith angles of 60° and 50° [Fig. 8(a) and (b)] shows that the BPRF increased with the decrease in BC content, such as snow with BC of 559 ng/g that had the largest BPRF, followed by snow with BC of 925 and 1356 ng/g. The effect of the change of BC content on the BPRF of snow can also be found at other viewing zenith angles in the forward scattering directions in the principal plane [see Fig. 8(c)]. This phenomenon can be explained as follows: with the increase of BC, the amount of single scattering of snow is reduced [29], and thus, a small BPRF corresponds to snow with high BC content. In the backward scattering directions [see Fig. 8(c)], the BC content has a weak effect on BPRF, such as the BPRF of snow with small BC that is slightly larger than that with high BC; however, we could not explain the reason for this behavior in this study.

3) Microscopic Roughness Effects: Roughness decreases the spectral BPRF of snow and disturbs the change of BPRF with viewing zenith angle in the forward scattering directions [see Fig. 9(a) and (b)]. Moreover, in the comparison between Fig. 9(c) and (d), it is found that roughness improves the BPRF in the backward scattering directions and changes the distribution of BPRF in other azimuth angles. In the forward

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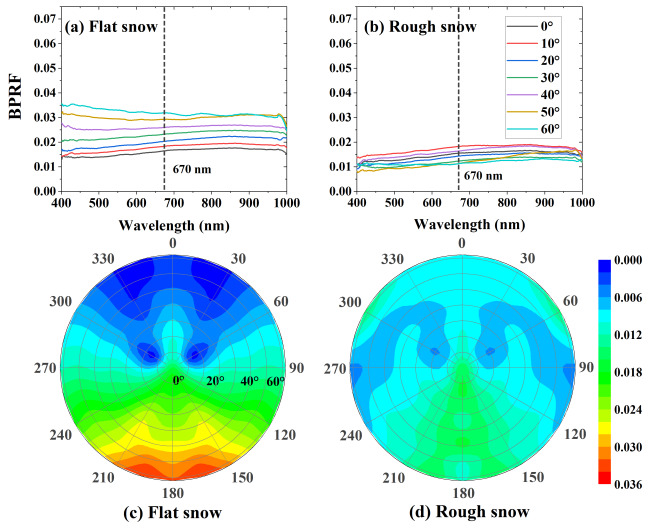


Fig. 9. Spectral BPRF of (a) flat and (b) rough snow surfaces measured at different viewing zenith angles in the forward scattering directions in the principal plane. The distribution of BPRF at 670 nm for (c) flat and (d) rough snow surfaces. The snow samples were measured on February 08, 2017 at Site 2 under artificial illumination.

scattering directions in the principal plane, the shadow of roughness can reduce the amount of single scattering light of snow [66], thus decreasing BPRF [see Fig. 9(b) and (d)]. In the backward scattering directions in the principal plane, compared with a flat snow surface, the enhanced multiple scattering of a rough snow surface may generate more negative polarization [80], which corresponds to a larger BPRF calculated by (7). These results indicate that microscopic roughness has a great effect on the BPRF of snow.

C. Photometric and Polarimetric Model Results

Snow optical models are necessary to simulate the reflection properties of snow because the input parameters related to snow are usually derived from modeled results in some climate models [25], [81]. In this section, both the $I\text{pRF}_{\text{lamp}}$ and BRF of snow are coupled with a photometric model to illustrate that the $I\text{pRF}_{\text{lamp}}$ is useful for inverting model parameters such as the BRF. The comparison between the measured and the modeled BPRF of snow was used to assess the ability of three polarimetric models in simulating the BPRF of snow.

1) Photometric Model Results: The ART model can simulate both spectral and angular reflection of snow with small amounts of pollution [9], [21], [27]. The BRF and $I\text{pRF}_{\text{lamp}}$ of new snow in the nadir direction were used to invert the ART model parameters, and the modeled spectral reflectance factors using the best inverted parameters (M and L) were compared with measured BRF and $I\text{pRF}_{\text{lamp}}$ [see Fig. 10(a) and (b)]. It is found that similar best inverted model parameters based on spectral BRF and $I\text{pRF}_{\text{lamp}}$ led to similar modeled reflectance factors and RMSE. In Fig. 10(a) and (b), the differences between modeled and measured reflectance factors around 2500 nm are due to the low sensitivity in the edge of the ASD spectrometer, which leads to the leap of the measurements around 2500 nm. This leap was also found in the previous studies under the lamp and sunlight conditions [10], [18]. The modeled angular reflectance factors (BRF and $I\text{pRF}_{\text{lamp}}$) of new snow using the best inverted model parameters are shown in Fig. 10(c) and (d). The similarity between modeled reflectance factors and the distribution pattern between modeled BRF and $I\text{pRF}_{\text{lamp}}$ indicate that the $I\text{pRF}_{\text{lamp}}$ has the same function as the BRF in simulating snow's photometric properties. These results in Fig. 10 further confirm that the $I\text{pRF}_{\text{lamp}}$ can replace the BRF of snow when the measured reflectance factors of snow were used to couple with a photometric model.

2) Polarimetric Model Results: The selected three polarimetric models have been used to simulate the polarization of land surfaces measured from airborne and spaceborne instruments [53]–[55]. However, these polarimetric models have not been used to model the BPRF of snow with different properties. In Figs. 11 and 12, the BPRF of new snow and old snow measured at different viewing zenith angles was used to compare with the modeled results from three polarimetric models. Both measured and modeled BPRF of snow have a forward scattering character, and they also have a similar distribution pattern. However, the M-model overestimates the BPRF of new snow in the forward scattering directions [located in the azimuth angles around 120° and 240° in Fig. 11(c)] and underestimates the BPRF of old snow in the forward scattering directions [Fig. 12(c)]. The large difference between measured and modeled BPRF (M-model) of new and old snow leads to larger RMSEs than that modeled using the NB-model and the L-model (see Table II). The RMSEs of new and old snow in Table II indicate that the L-model has the best predicated BPRF, followed by the NB-model and the M-model. This may be explained as follows. All the three models are based on the Fresnel equation, the L-model has three free parameters, the NB-model has two free parameters, the M-model has one free parameter, and the greater the number of free parameters, the better the agreement between model and observations. In this study, we did not discuss the theory of the three selected polarimetric models; the models were chosen to simulate the BPRF of land surfaces based on their advantages [54], [55], [60], [61].

Unexpectedly, the BPRF of rough snow at 670 nm [Fig. 13(a)] does not have a similar distribution pattern as the BPRF modeled using the three polarimetric models [Fig. 13(b)–(d)]. The minimum measured BPRF of rough snow is found in the directions perpendicular to the principal plane, and the BPRF is relatively large in the backward scattering directions [Fig. 13(a)] compared with that of flat snow [Figs. 11(a) and 12(a)]. However, the polarimetric models based on the Fresnel equation can only generate a minimum BPRF in the backward scattering directions [60], [82]. It is difficult to calculate the similar BPRF in the hemispherical space as the measured results shown in Fig. 13(a). The polarimetric model results basically coincide with the measured BPRF of snow with different properties (except for microscopic roughness) in the hemispherical space [see Figs. 11–13 and Table II]. New polarimetric models will be needed to develop in future studies to simulate the BPRF of rough snow.

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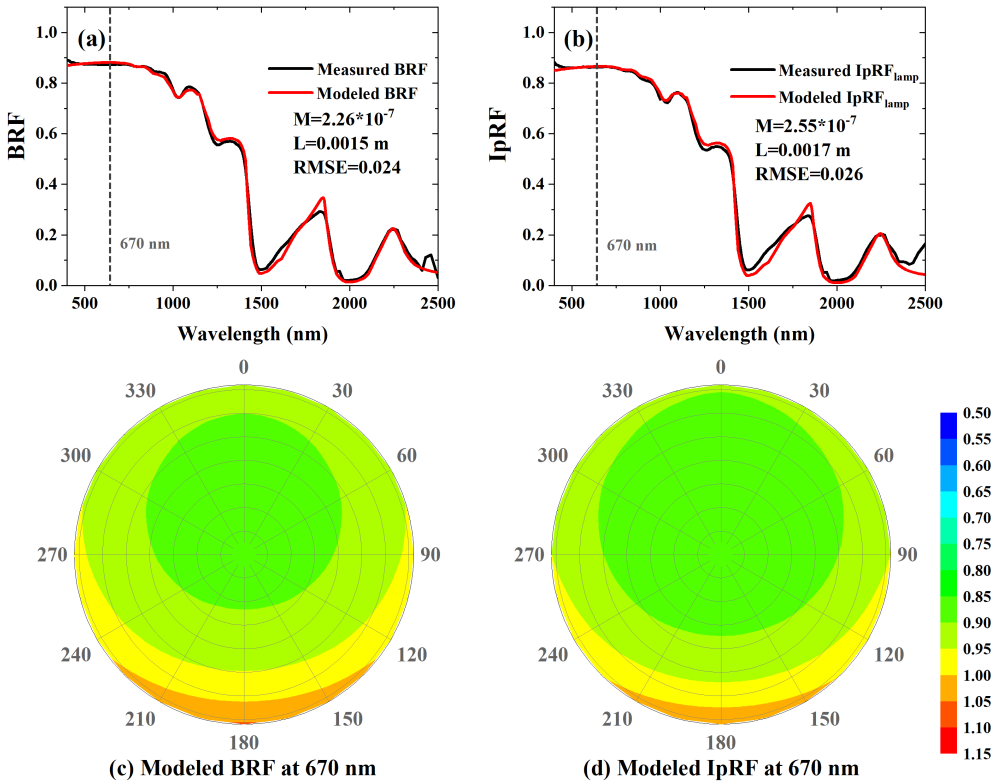


Fig. 10. Comparison between measured and modeled spectral (a) BRF and (b) IpRF$_{\text{lamp}}$ of new snow (February 10, 2017, at Site 2) in the nadir direction using the ART model in (11). (c) and (d) Modeled angular BRF and IpRF$_{\text{lamp}}$ of new snow at 670 nm using the ART model.

TABLE II

ROOT MEAN SQUARE ERRORS (RMSES) OF SNOW WITH DIFFERENT PROPERTIES AT 670 nm FROM THREE POLARIMETRIC MODELS

Snow samples	M-model	NB-model	L-model
New snow	0.0040	0.0022	0.0021
Old snow	0.0036	0.0017	0.0014
Rough snow	0.0055	0.0036	0.0025

IV. DISCUSSION

A. Similarity Between BRF (or HDRF) and $IpRF_{\text{lamp}}$ (or $IpRF_{\text{sun}}$) of Snow

Both intensity and polarization are useful for understanding the optical properties of light reflected from snow. Therefore, the intensity is usually derived from photometric measurements [36], [83], while the polarization is usually obtained from polarimetric measurements [53], [52]. The two different measurement processes complicate the characterization of the optical properties of snow with different properties. In theory, the Stokes parameters can be used to describe both intensity and polarization properties of light reflected from land surfaces: the I parameter represents the reflected intensity, while Q and U represent the polarization [41]. Yet, the reflected intensity taken from polarimetric measurements (IpRF$_{\text{lamp}}$ or IpRF$_{\text{sun}}$) had not been confirmed to be similar to that taken from photometric measurements (BRF or HDRF). In this study, we found a good agreement between IpRF$_{\text{lamp}}$ (or IpRF$_{\text{sun}}$) and BRF (or HDRF) of snow with different properties and then confirmed that the IpRF$_{\text{lamp}}$ (or IpRF$_{\text{sun}}$) is useful for relating snow properties and inverting photometric model parameters as the BRF (or HDRF). Thus, polarimetric measurements (considering the absorption ratio of the polarizer) obtain both the intensity and the polarization of snow. This is useful and convenient for deepening our understanding of the optical properties of light reflected from snow.

B. Understanding the Effects of Snow Properties on BPRF

Several researchers have used the degree of linear polarization (Dolp to characterize the polarization properties of light reflected from snow surfaces [49]–[51], [52]. However, the Dolp $= R_p/I$ is the combination of polarized and total intensity, and thus, it represents a part of the polarization properties of snow. Moreover, most of the polarized reflectance models used to calculate the polarization of land surfaces cannot directly couple with the Dolp [47], [54], [60], [82]. BPRF is only related to the polarization properties of snow, and several polarimetric models only simulate the BPRF. Thus, it is necessary to investigate the effects of snow properties on

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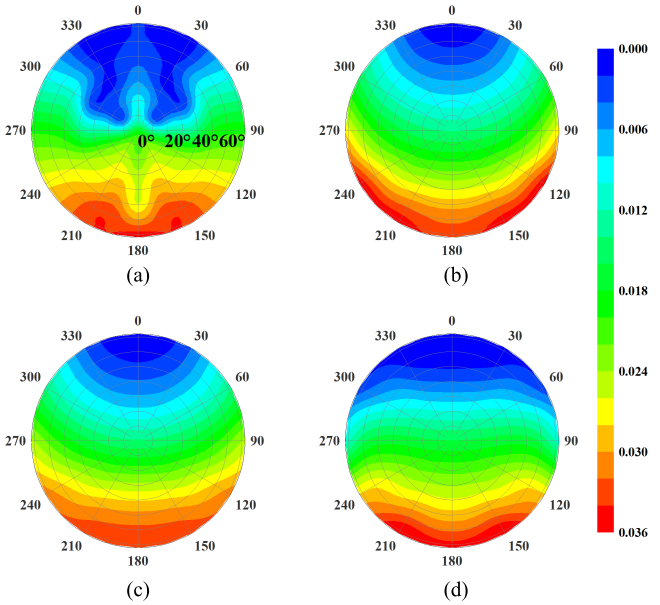


Fig. 11. (a) Measured BPRF of new snow at 670 nm at Site 2 on February 10, 2017. Modeled BPRF at 670 nm using (b) M-model, (c) NB-model, and (d) L-model.

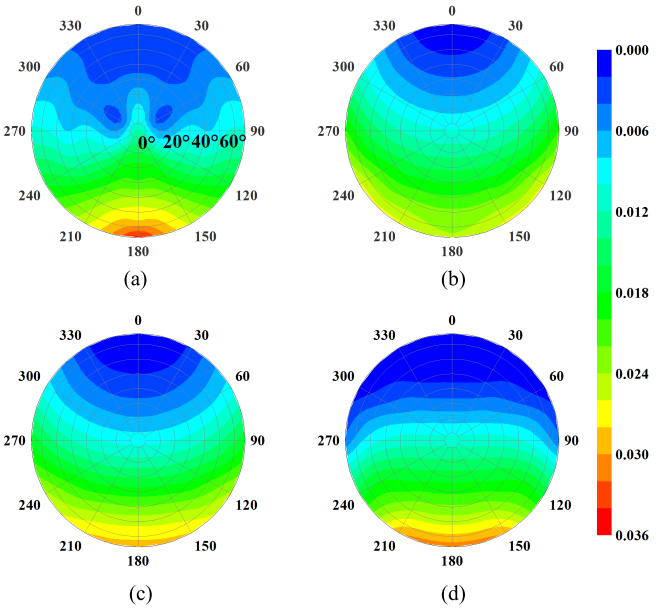


Fig. 12. (a) Measured BPRF of flat old snow at 670 nm at Site 2 on February 6, 2017. Modeled BPRF at 670 nm using (b) M-model, (c) NB-model, and (d) L-model.

BPRF, which can deepen our understanding of the relationship between optical properties and snow properties.

Changing snow properties equally influence the BPRF and its distribution at each wavelength, especially in the forward scattering directions. This is because the BPRF is weakly wavelength-dependent and has a forward scattering character [60], [82]. In this study, assuming that the single scattering light of snow is the source of BPRF, the BPRF can be related to the snow properties. Under this assumption, the decrease

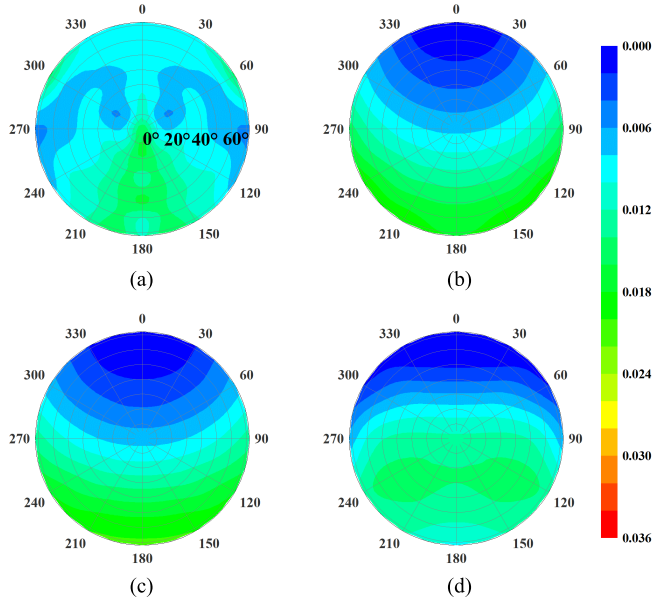


Fig. 13. (a) Measured BPRF of rough snow at 670 nm at Site 2 on February 6, 2017. Modeled BPRF at 670 nm using (b) M-model, (c) NB-model, and (d) L-model.

of grain size and the increase of pollution and microscopic roughness lead to a reduction of single scattering light and then a reduction in BPRF in the forward scattering directions. However, this study could not explain the effects of snow properties on BPRF in the backward scattering directions. In these directions, the BPRF is small, and thus, it is difficult to ensure that the change of BPRF is caused by the variation of snow properties. On the other hand, several physical mechanisms contribute to the polarization of snow in the backward scattering directions [50], [80]. More polarimetric measurements of snow with different properties are needed to assess the dependence of the BPRF on the snow properties in the backward scattering directions.

The shape of the microscopic roughness in this study differs from that in previous studies [65], [73], but the enhanced reflectance factors (in the backward scattering directions) and the reduced reflectance factors (in the forward scattering directions) characteristics are similar to previous measured and modeled results [65], [66], [73], [74]. The reflectance factors and BPRF of the rough snow surface were taken under the same measurement conditions; thus, the special distribution pattern of the BPRF of a rough snow surface (with a relatively low BPRF near the directions perpendicular to the principal plane and the minimum BPRF in the backward scattering directions) can reliably represent the polarization properties of rough snow. This study only indicated that the polarization depends on the snow properties and that the BPRF is required for describing the optical properties of light reflected from different types of snow. In future studies, the shape of snow grains, the kinds of pollution, and the pattern and intensity of roughness should be considered [66], [72] because these properties may also influence the magnitude and distribution of BPRF.

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C. Need for Polarimetric Model Simulations of Snow

Photometric models have been widely used to simulate the spectral and angular reflection of snow with different properties based on spaceborne, airborne, and ground measurements [13], [29], [35], [84]. These modeled results have been used for global, regional, and plot studies related to snow [32], [37], [81]. However, the BPRF of snow cannot be separated from the reflection calculated by the photometric models. With advances in polarimetric instruments, the BPRF of snow has been measured from spaceborne, airborne, and ground platforms [47], [53], [54], [56]. These polarization measurements of snow were coupled with polarimetric models to deepen the understanding of snow's optical properties and improve the retrieval of snow grain size when combining with photometric results [50]–[53]. Moreover, the polarimetric model results of snow should be separated when the polarization is used to retrieve aerosol properties [54], [56]. Thus, it is necessary to calculate the BPRF of snow with different properties using polarimetric models.

Generally, polarimetric models generate a large BPRF in the forward scattering directions and a small BPRF in the backward scattering directions. This BPRF distribution pattern can be explained by the Fresnel equation and is representative of the distribution from flat snow surfaces with different properties (such as with different grain sizes and different pollution contents). For rough snow surfaces, the polarimetric models did not generate a BPRF distribution pattern similar to the measurements. Similar to the improvements in the photometric models for the calculation of BRF or HDRF [36], [65], [73], [74], the polarimetric models should also be improved to accurately simulate the BPRF of rough snow in future studies. This will help us understand the relationship between the polarization and the microscopic roughness of snow.

V. Conclusion

In this study, multiangular photometric and polarimetric hyperspectral measurements of snow with different properties (grain size, pollution level, and microscopic roughness) were first performed under sunlight and artificial illumination conditions. Then, the intensity signals derived from photometric (BRF or HDRF) and polarimetric ($I\mathrm{pRF}_{\mathrm{lamp}}$ or $I\mathrm{pRF}_{\mathrm{sun}}$) measurements were confirmed to be equally related to snow properties and coupled with photometric models when the absorption ratio of the polarizer was considered. Moreover, the polarization property (defined as BPRF) of snow was found to depend on the snow properties: the smaller the grain size, the larger the BPRF; the larger the amount of pollution, the smaller the BPRF; and snow with microscopic roughness generates smaller BPRF than flat snow surfaces. These polarization properties can be explained by the change of single scattering light from different types of snow surfaces because single scattering light is the base of the BPRF of snow. However, in this study, we could not establish a relationship between BPRF and snow properties near the backward scattering direction in the principal plane.

The three selected polarimetric models did not have a similar ability for simulating the measured BPRF of flat snow surfaces. We found that a greater number of free parameters in the model resulted in a better agreement with measured BPRF. Unfortunately, these polarimetric models cannot calculate the BPRF of snow with microscopic roughness over the hemispherical space, especially when the BPRF is measured in the directions perpendicular to the principal plane and in the backward scattering directions.

Overall, polarimetric measurements can obtain both the intensity ($I\mathrm{pRF}_{\mathrm{lamp}}$ or $I\mathrm{pRF}_{\mathrm{sun}}$) and the polarization (BPRF) of snow. The former has a similar response to snow properties as the photometric measurements; the latter is also sensitive to snow properties, which can be mostly explained by physical mechanisms. These results indicate that the polarimetric measurements of snow are useful for studying snow properties, allowing, on the one hand, to deepen the understanding of the optical properties of light reflected from snow, and on the other hand, combining the intensity and polarization measurements may improve the estimation of snow properties. Although this study improved our understanding of the relationship between BPRF and snow properties, the polarization of snow with different properties in the backward scattering directions could not be explained, and the polarimetric models could not model the BPRF of snow with microscopic roughness. Thus, future studies should focus on polarimetric measurements and models of snow.

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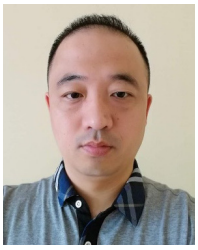
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